

Before I explain it, let me first close off the open points from our previous discussion.

STM32 and WiFi — No Single Chip Solution

As I mentioned earlier, STM32WB55 supports BLE only and aligns well with your BLE preference. After further research, I can confirm that the STM32 family does not offer a single chip solution combining motor control capabilities with integrated WiFi. Any WiFi-based approach would require an external module, adding hardware complexity and cost. Since your preference is BLE, this is not a concern — but it does rule out Mongoose in its standard form, as Mongoose requires a TCP/IP (WiFi) connection to serve its web interface.

Why Mongoose Does Not Work With BLE

Mongoose is a web server framework that operates over TCP/IP. The browser-based dashboard you saw in the Mongoose documentation is only possible when the device is connected via WiFi. There is no way to run Mongoose over a BLE connection regardless of the architecture.

The Architecture That Solves This — Web Bluetooth

You asked whether a different architecture could enable a browser-only, no-install solution over BLE. The answer is yes.

The approach is called Web Bluetooth. It is a modern browser capability that allows a web page running in Chrome to connect directly to a BLE device — with no backend service, no installed software, and no app store. This directly addresses the concern you raised about our Option 1 requiring a background service on the PC.

The solution works as follows. The dashboard is hosted online as a web page at a fixed URL — similar in concept to what you saw with Mongoose, but without requiring WiFi on the device. When a user wants to interact with the motor controllers, they simply open that URL in Chrome. The browser then connects directly to the STM32WB55 over Bluetooth. No installation of any kind is required on the user's device.

From the user's perspective:

- They open a link in Chrome, exactly like any website
- The browser asks permission to connect to the Bluetooth device
- The dashboard shows live speed curves, cumulative rotations, elapsed cycle time, and all settings
- Parameters can be changed and commands sent directly from the browser
- Nothing is installed on their machine

This also handles your modularity requirement cleanly. The browser can maintain simultaneous BLE connections to all four STM32WB55 controllers from a single tab, displaying each module in its own panel in one unified dashboard — matching your Scenario 1 from the specification.

Platform Compatibility

This solution works seamlessly on:

-  Windows (Chrome or Edge)
-  Linux (Chrome)
-  Android phones and tablets (Chrome)
-  macOS — Apple laptops and desktops (Chrome)
-  iOS — iPhone and iPad

I want to be transparent about the iOS limitation. Apple currently does not allow browsers on iPhone and iPad to access Bluetooth. This is an Apple platform restriction and cannot be worked around without a dedicated iOS application. If iOS support is a firm requirement, we would need to discuss a separate native app for that platform, which would involve additional effort. If iOS is not critical for your users, all other platforms are fully covered with no installation needed.

Proposed Hardware

For the controller, we recommend the STM32WB55 — staying within the STM32 ecosystem as you prefer, with built-in BLE 5.0 and strong motor control peripherals, with no external wireless module required.

Please let us know whether iOS support is a requirement for your users, and we will finalize the architecture and move forward with the detailed proposal.

Best regards,

Khaled

Sunday, Mar 29

Khaled Hassanain

3:35 PM

Hi Toon,

One quick update on the iOS question — I found a clean solution.

On iOS, Web Bluetooth requires using a free browser called Bluefy (available on the App Store). It's a one-tap install, works exactly like installing Spotify. Once installed, the dashboard opens and works normally — no extra steps.

This means our Web Bluetooth dashboard works on:

- Android → Chrome (built-in)
- Desktop → Any browser
- iOS → Bluefy (free, one-tap)

No additional hardware or backend needed. I think this completes our solution.